Expt. 2 - γ absorption. (D)

5.2.2 BASIC KEY FUNCTIONS

Following table shows the Basic Keys Symbol and their function. These key symbols are used for displaying the [illegible]

Key Symbol	Functions
S	SKIP KEY for changing Operating Mode or jump to next Mode
B	BACK KEY to go back without making any changes
E	ENTER KEY to confirm the parameter entry
+	INCREMENT KEY to increment the digit under cursor
>	SHIFT KEY to move the cursor to next digit position
II	STOP or HALT KEY

Apart from these Basic Key functions, there are many other key symbols used for displaying the options. These key symbols are indicated while describing the particular parameter/operation.

5.2.3 SET PARAMETERS

In this mode, user can set the parameters such as Preset Time, Number of Runs etc. Counts per Run are stored automatically in the memory and can be retrieved in RECALL MODE.

While the Menu is indicating

```
Para Electronics
NUCLEAR COUNTER
```

After few seconds

```
GM Counter
Model PNC-1G
```

After few seconds

```
Set Time / Runs?
T U D S  Default
```

After few seconds

```
HV = 0050 Volts
Press Any Key ...
```

After pressing any key

```
HV Read Mode
E S
```

Press → [illegible] using Helipot dial on the front panel

```
HV  =  0052  Volts
Press Any Key ...
```

Press any key to proceed

```
Set Time / Runs?
T U D S  Default
```

Press:- → **T** for Set Time **U** for RUns?
In this setup user can set Time and Runs

T Key: - Set Preset Time.
U Key: - Set Number of Runs
D Key: - Set Default Parameters
S Key: - Go to Back

If **T** Key is pressed, following menu will appear on LCD

```
Counting Time
+ >  E  B  0000 Sec
```

Counting Time can be set between 1 to 9999 Seconds.

If **U** Key is pressed, following menu will appear on LCD

```
No. Of Runs  0010
+ >  E  B
```

Number of Runs can be set between 1 to 1000.

Note:

Menu indicates blinking underlined digit that can be modified. To increment this digit, press + Key, to shift cursor to next digit press > Key E Key to accept the set parameter and B Key to exit from this Mode.

If Counting Time or Number of Runs is set to 0000, PNC-1G will load Counting Time as 1 Second and Number of Runs as 1.

Press → **D** Key: Default Parameters will be loaded.

PNC-1G will prompt on display Load default Parameters.

Preset Time : - 1 Second
Number of Runs : - 10

5.2.4 RUN MODE

S Ke[illegible] as below

Start Counter?
▶ S

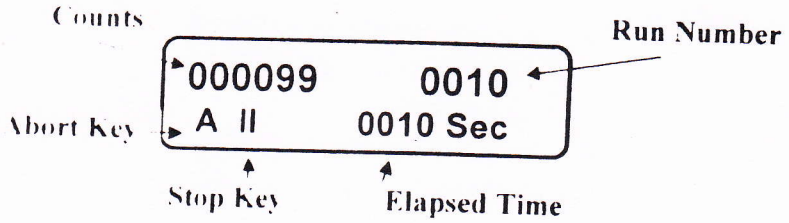
Press ▶ se[illegible] t[illegible] tart counter. Following message will appear on the LCD for few seconds before actual counting starts.

Paralysis Time
U E S 250u S

*(** Note :- 'Paralysis Time' option is provided only if it is ordered)*

During counter initialization, paralysis time can be set by the user just before counting starts.

While counting, Counting Time, Number of Runs and counts as shown on LCD display. All these three values of the current run are stored in EEPROM. The maximum storage is upto 1000 Runs.



A Key :- Abort key will terminate the current Run and halt counting immediately. The data of the current Run is ignored by **PNC-1G**

A Run can be halted / stopped by pressing **II** key. However, the stopping will occur only when current Run is completed. On halt, instrument will prompt user whether to continue or to start fresh new run.

C Key :- To continue run from where the run was stopped
▶ Key :- To start run from fresh
S Key :- To change Mode

When preset numbers of Runs are completed the menu will change to

000099	0010
▶ S	0010 Sec

► Key [illegible] fresh.
S Key [illegible] Mode

[illegible] before preset number of runs are completed, it is possible to [illegible] continue from the next run where the stop as occurred

```
000099        0010
 C ►  S    0010 Sec
```

► Key [illegible] fresh.
C Key - To continue from where stopped.
S Key - To change Mode

*** Note - Due to overheating of electronic components, counter - timer may misbehave, in that case please Turn OFF the instrument for few seconds and then restart it. Set HV as suggested, wait for a minute till High voltage stabilizes and start counting experiment*

5.2.5 RECALL DATA MODE

In this mode the Counts Data (as stored in Preset Timer Run) can be retrieved and transferred to LCD or COM Port or Printer Port. It also allows user to select the data range which will be transferred to the set device.

Press 2 S Key to get this mode. Menu will indicate

```
Recall Data?
  E  S
```

Press **E** - To recall Counts Data
Press **S** - To go back to earlier menu

Pressing of **E** key will prompt the selection of Data Range which can be Entire or Partial.

```
Data From / To
F  T  E  B    Entire
```

F Key - To select data "From Run" (beginning of data range)
T Key - To select data "To Run" (end of data range)
E Key - To select Entire data range (all the Runs)
B Key - Back to Recall Data

It is necessary to enter valid range of data. PNC-1G will not accept invalid range.

Next selection is for the Device where the data is to be transferred, i.e. either on LCD display or on Printer or on COM port for transfer to PC.

```
Data ON  LCD
L  P  B      Printer
```

L Key - To view data on LCD display.
P Key - To transfer data on any standard Printer.
B Key - To go back to earlier menu

Run PNC-1G data transfer program on PC. Click on 'Receive data' button to receive all data from the instrument to PC. User can not receive data when instrument is counting.